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PARASITOLOGICAL DIAGNOSIS OF SLEEPING SICKNESS

Veerle Lejon, Philippe Büscher

Parasitological diagnosis is achieved by microscopic examination of lymph node aspirate, blood or cerebrospinal fluid. It gives direct evidence for trypanosome infection and thus definite diagnosis. Parasite detection can be labor-intensive. In practice, parasitological examination is therefore limited to clinical suspects for *Trypanosoma brucei rhodesiense* infection and/or serological suspects for *Trypanosoma brucei gambiense* infection. Techniques are poorly sensitive and in use for many decades. Parasite loads can vary between more than 10,000 trypanosomes/ml, being easily detectable, and less than 50 trypanosomes/ml, particularly in *T.b. gambiense* infection, being below the detection limit of the most sensitive methods. Failure to demonstrate parasites therefore does not exclude infection. Repeated examination on consecutive days can increase test sensitivity but in practice is seldom done. It is recommended to use methods that allow examination of larger sample volumes. Concentration or cultivation techniques often combined with optimized visualization systems can also increase the sensitivity of parasitological diagnosis.

For efficient parasitological examination, it is important to keep the time between sampling and examination as short as possible to avoid immobilization and subsequent lysis of trypanosomes in the sample. Trypanosomes are rapidly killed by sunlight but survive longer when the sample is kept cool in a dark place. Special attention should be paid to the quality of reagents and materials and to proper maintenance of the equipment, in particular the microscope. In general, a 10x10 or 40x10 magnification is used for screening the sample under the microscope.

Hereafter, an exhaustive list of trypanosome detection techniques is described of which only some are applied for diagnosis of sleeping sickness in HAT treatment centers. Several reviews deal with the techniques in detail [1-4].

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PARASITOLOGY ON CHANCRE AND LYMPH NODE ASPIRATES

Chancre aspirate.

Trypanosomes can be detected in the chancre a few days earlier than in the blood. The chancre exudate is microscopically examined as a fresh, fixed or Giemsa stained preparation.

This technique is the earliest way to diagnose trypanosome infection [5], and is simple to perform, cheap, and gives a rapid result. However, it is seldom applied since the chancre sometimes remains absent and in most of the patients has already disappeared at time of examination. The chancre is more commonly observed in *T.b. rhodesiense* than in *T.b. gambiense* infections and more Europeans, whereas only rarely in Africans.

Lymph node aspirate.

When enlarged cervical lymph nodes are present (Fig. 1), they can be punctured and the fresh aspirate expelled onto a microscope slide. A cover slip is put on to extend the sample and facilitate the reading. The fresh preparation is then examined microscopically (40x10) for the presence of motile trypanosomes.



Figure 1 - Lymph node palpation. If enlarged lymph nodes are present, they can be punctured.

Due to its simplicity and low cost, this technique is widely applied. Sensitivity varies from 40%-80% and seems to depend on the parasite strain, on the occurrence of other diseases causing lymphadenopathy and on the disease stage (cervical lymph nodes are more common in early stage).

In mass screening for *T.b. gambiense* infection, the cost effectiveness of cervical lymph node palpation as a first line examination is questioned [6, 7]. For example, in CATT negative individuals in the Democratic Republic of Congo between 1999 and 2001, trypanosomes were observed in only 316 (0.18%) out of 174,295 lymph node aspirates. Inversely, all patients with positive lymph node aspirates found among 1000 individuals from two endemic foci in Angola and Central African Republic had a positive CATT [6].

PARASITOLOGY ON BLOOD

Wet blood film.

In wet blood films, about 5-10 μl of finger prick blood are applied on a microscope slide and examined microscopically under a cover slip. If present, trypanosomes are revealed by their movement between the red blood cells (the agitation of surrounding erythrocytes often attracts the attention).

The detection limit in wet blood films is 10 000 trypanosomes/ml, corresponding to 1 parasite/200 microscopic fields. Sensitivity is thus low [8].

This technique is not optimal for detection of *T.b. gambiense*, but still in use, especially in *T.b. rhodesiense* areas, because of its low cost, simplicity and immediate result. Giemsa or Field stained thin blood films have a similar low sensitivity.

Thick blood film.

The thick blood film is used for diagnosis of both *T.b. rhodesiense* and *gambiense* disease. A small drop (20 μl) of finger prick blood is spread over an area of 1 cm^2 on a microscope slide and defibrinated with a tooth pick or the corner of another slide. The slide is dried in a horizontal position, protected from direct sunlight, and is stained by Giemsa or Field stain prior to microscopic examination at 40x10 or 100x10 magnification.

The detection limit of the thick blood film is in practice only about 5000 trypanosomes/ml.

Thick blood film examination is relatively time consuming (10-15 minutes per slide). Proper preparation of the sample is important to avoid artifacts: correct thickness and defibrination, purity and quality of the staining reagents and

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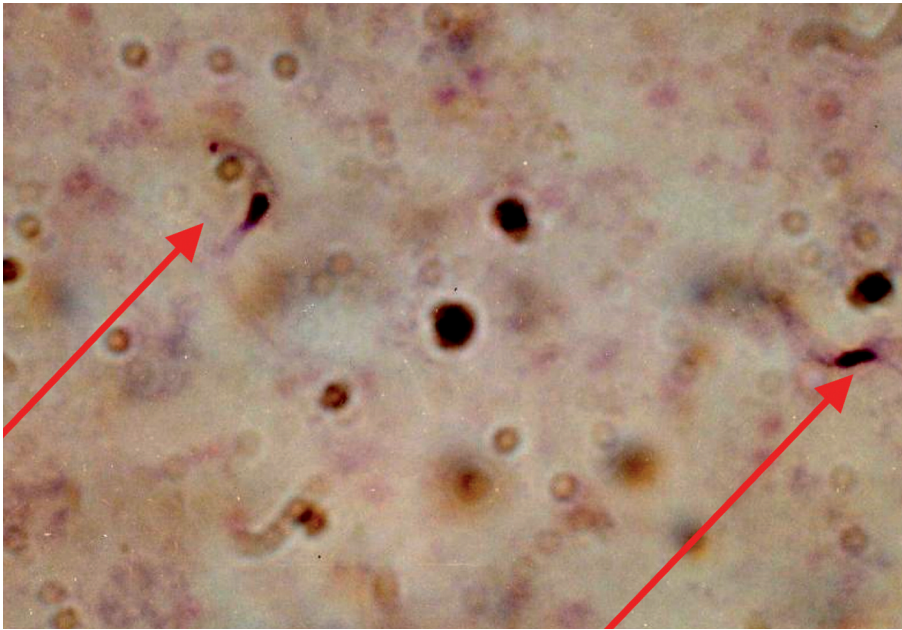


Figure 2 - Stained thick blood film viewed under a 100x10 magnification.
Two trypanosomes are visible (arrows).

cleanness of the slides. Often serious deformation of trypanosomes can occur. Before a sample can be considered positive, the microscopist should pay attention to the morphology of the trypanosomes and recognize at least 4 out of 5 characteristic criteria: the stained nucleus, the stained kinetoplast, the stained flagellum, the cell body and the correct size (Fig. 2).

In those situations where a centrifuge is not available, thick blood film is the technique of choice for blood examination: it is simple to perform and cheap. Apart from trypanosomes, other parasites such as microfilaria and *Plasmodium* can be detected.

Micro haematocrit centrifugation technique (mHCT).

This blood concentration technique, also called the «capillary tube centrifugation technique (CTC)» or «Woo test », was developed more than 30 years ago and is still in use in many HAT control programs [9-11]. Capillary tubes containing anticoagulant are filled for three quarters (about 50 µl) with finger prick blood. The dry end is sealed with plasticine or is carefully flame sealed using an alcohol burner (Fig. 3a, 3b). By high speed centrifugation in an haematocrite centrifuge, trypanosomes are concentrated at the level of the white blood cells, between the plasma and the erythrocytes. The capillary tubes are mounted in a

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holder or between a microscopic slide and a coverslip. The empty space between the glass surfaces is filled up with water to reduce diffraction. Capillaries are directly examined at low magnification (10x10 or 10x20) for mobile parasites. It is recommended to turn the capillary over 360° during examination.

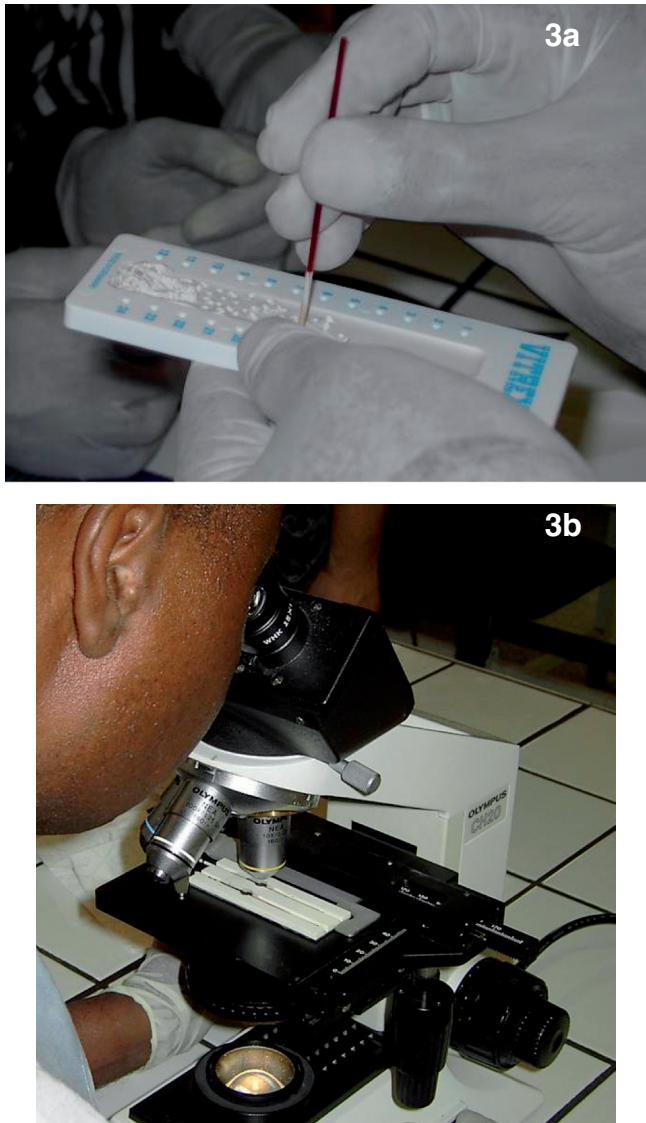


Figure 3 - Micro haematocrite centrifugation technique. Capillary tubes are filled for three quarters and the dry end is sealed with plasticine (3a). After centrifugation, the capillary tubes are mounted in a holder and the buffy coat is examined at 10x10 magnification (3b).

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The detection limit of mHCT is estimated to be around 500 trypanosomes/ml.

To increase sensitivity, examination of up to 8 tubes per person has been recommended.

The mHCT is cheap, moderately time-consuming and has acceptable sensitivity. It is a relatively simple technique that can be applied during mass screening by mobile teams.

Disadvantages of the mHCT are the need for a micro haematocrite centrifuge and electricity (generator for mobile teams) and the need of some experience to read the result, since trypanosomes appear very small, and are present between white blood cells. Unfortunately, the presence of moving microfilariae in the blood often makes it impossible to observe trypanosomes.

Quantitative Buffy Coat.

Acridine orange is a fluorescent dye that binds to DNA in the nucleus and the kinetoplast of living trypanosomes in fresh blood, thus allowing discrimination from white blood cells in which only the nucleus becomes fluorescent. The Quantitative Buffy Coat (QBC), initially developed for the rapid assessment of differential cell counts, has been extended to the diagnosis of haemoparasites including *Plasmodium* and *Trypanosoma* [12, 13]. QBC has been used with success in sleeping sickness diagnosis [13] and combines the fluorescent technique with the concentration of parasites by centrifugation. After high speed centrifugation of the blood in special capillary tubes, coated with EDTA and acridine orange and containing a small floating cylinder, mobile trypanosomes can be identified by their fluorescent kinetoplast and nucleus between the white blood cells in the expanded buffy coat. UV light is generated by a 'cold light source' connected by a glass fiber to a special objective containing the appropriate filter. This objective can be mounted on almost every microscope.

The QBC is a sensitive technique that allows the diagnosis of concomitant malaria, which is very useful for patients' care. With a detection limit of <500 trypanosomes/ml, the QBC can detect more patients with low parasitaemia than the mHCT when less than 8 capillary tubes are used. It seems to be almost as sensitive as the Mini Anion Exchange Centrifugation Technique (mAECT) described below [14, 15]. However, as in mHCT, the detection of trypanosomes in QBC becomes very difficult when microfilaria are present.

The technique is relatively costly: an initial investment into a special centrifuge and light source is needed, and capillaries cost 1.5-3 US\$. The relative sophistication and fragility of the material prevents its daily transport during active screening sessions. A dark room is needed for reading the result. The pro-

duction of the QBC kit has been abandoned for some time, but the manufacturing of capillaries seems to have resumed (QBC Diagnostics Inc., Beckton-Dickinson).

Mini-Anion-Exchange Centrifugation Technique (mAECT).

The mAECT has been introduced by Lumsden et al. [16], based on a technique developed by Lanham and Godfrey [17]. At a pH of 8, trypanosomes are less negatively charged than blood cells. They can be separated from venous blood by anion chromatography using DEAE cellulose. The red blood cells stay on the DEAE gel, and the eluent containing the trypanosomes is collected in a sealed tube. Trypanosomes are concentrated at the bottom of the tube by low speed centrifugation (3000 rpm). The tip of the tube is then examined in a special holder under the microscope (10x10) for the presence of trypanosomes (Fig. 4).

The large blood volume (up to 500 μ l) enables detection of less than 50 trypanosomes/ml resulting in high diagnostic sensitivity. In case of low parasitaemia and failure of all other techniques to detect the parasite, mAECT can make the difference. If microfilariae are present, they might also migrate through the column but they seem to interfere less than in the mHCT (J. Kwete, personal communication).

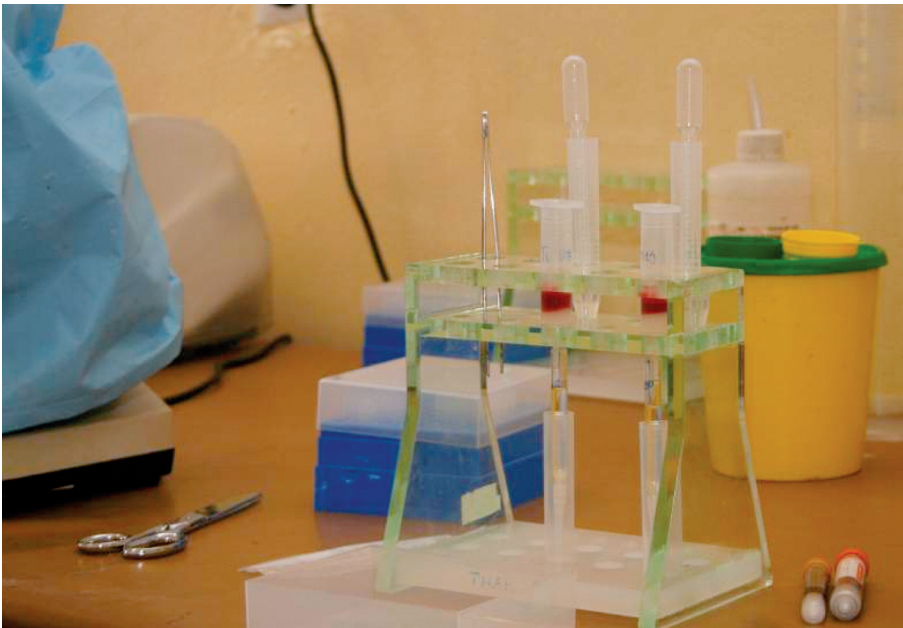


Figure 4 - The mAECT.. Red blood cells stay on the DEAE gel, the eluant containing trypanosomes is collected in a sealed tube.

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The manipulations to perform a mAECT are quite tedious and time-consuming and need for a table centrifuge renders the implementation of the mAECT during mobile screening difficult. The price of 3 Euro may be prohibitive for large scale use.

mAECT columns and accessory materials are available from Projet de Recherches Cliniques sur la Trypanosomiase (PRCT, BP 1425, Daloa, Côte d'Ivoire). A new simplified model is being produced at Institut National de Recherches Biologiques (INRB, Avenue de la Démocratie, Kinshasa, République Démocratique du Congo).

In vitro culture.

Due to low success rates, in vitro culture is seldom applied for diagnosis. The Kit for In Vitro Isolation (KIVI) was originally developed for isolation of trypanosomes under field conditions [18] but can be used for diagnosis when other techniques fail. The technique consists of inoculating aseptically a large volume of venous blood (10 ml) into culture medium whereafter bloodstream trypanosomes transform into procyclics and start to multiply. The culture flask is kept in the dark at ambient temperature (ideally 27°C) and should be examined for the presence of trypanosomes during several weeks.

Xenodiagnosis.

Xenodiagnosis can be performed by inoculating rodents with the blood of the putative host or by feeding teneral tsetse flies on the host. Xenodiagnosis is seldom applied due to the need of laboratory animals, the low success rates and long incubation time.

CEREBROSPINAL FLUID

Examination of the CSF is usually undertaken for stage determination and follow-up, and not for primary diagnosis, except in persons with strong clinical or serological suspicion on whom trypanosome detection in blood or lymph failed (see chapter Stage determination and follow-up).

SENSITIVITY AND DETECTION LIMIT OF PARASITOLOGICAL TECHNIQUES

One has to discriminate between analytic sensitivity, the number of trypanosomes that can be detected per unit volume (detection limit here after) and diagnostic sensitivity, the number of patients that can be detected with a certain technique (sensitivity here after).

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The detection limit of a test is mainly defined by the volume of sample that is examined. Based on the sample volume a theoretical detection limit can be calculated (Table 1) but differs significantly from the detection limit achieved in practice. Other parameters that may influence the detection limit are visibility of trypanosomes in the applied technique, presence of microfilariae, time of reading (determines for example the number of fields that will be examined) or the attention of the microscopist [19].

Few authors have studied the comparative diagnostic sensitivity of several tests in a standardized way and on large patient numbers. As visible from table 2, the percentages should be interpreted as relative, rather than absolute.

Table I - Relationship between sample volume and detection limit of parasite detection techniques on blood

Technique	Sample volume (μl)	Theoretical detection limit Trypanosomes/ml	Practical detection limit Trypanosomes/ml
Wet blood film	5-10	100	6000-10000
Thick blood film	25	40	600-5000
mHCT	50	20	500-600
QBC	50	20	15-300
mAECT	500	2	15-50

Table II - Comparison of percent sensitivity of parasitological techniques according to different authors.

- a) 26 measurements on 6 trypanosome positive individuals;
 b) 35 mAECT positive patients;
 c) 9 trypanosome positive patients, some were retested, 1 capillary/mHCT;
 d) 13-49 trypanosome positive patients tested;
 e) 12-17 trypanosome positive patients tested;
 f) 154 trypanosome positive patients tested, 4 capillaries/mHCT

	a	b	c	d	e	f
Reference	20	20	21	22	23	24
Lymph aspirate				59%	63%	19%
Wet blood film			25%	22%		4%
Thick blood film	30%	63%	56%	35%	100%	27%
mHCT	46%	73%	13%	48%	44%	57%
QBC					93%	
mAECT	88%	100%	54%	85	80%	75%
KIVI					94%	
Rat inoculation of blood			57%	31%		

MOLECULAR DIAGNOSIS

Polymerase chain reaction (PCR) is a molecular technique that can detect trypanosomal DNA and can be considered as a surrogate for parasite detection (Fig. 5). PCR can be applied on any patient that may contain parasite DNA, such as whole blood, buffy coat, lymph node aspirate, or CSF (for PCR on CSF see chapter 5.4: Stage determination and follow-up). Unlike parasitology, the method provides indirect evidence of infection and additional research is still necessary to render PCR a real tool for diagnosis and control of human African trypanosomiasis [25].

Most diagnostic PCR studies are based on primers derived from satellite sequences or multi-copy genes. Through targeting repetitive sequences lower detection limits and higher diagnostic sensitivities can be obtained compared to targeting low-copy or single copy sequences.

Often used is the 177 bp repetitive satellite DNA sequence of the *Trypanozoon* group [26-29], using the TBR primer set [30], or other regions within the same repeat [31]. Using these primers, PCR claims i) to detect between 1 and 25 trypanosomes/ml [26, 28] and ii) to be systematically more sensitive than parasitological examination with sensitivities of 99.4-100% and being able to detect trypanosome nucleic acid in part of the parasitologically unconfirmed CATT seropositive suspects [26, 27].

The expression site associated genes (ESAG) 6-7 sequence, is a multi-copy gene specific for the *Trypanozoon* group [32]. Using a primer set targeting ESAG 6-7, a comparable detection limit of 40 trypanosomes/ml is obtained together with a diagnostic sensitivity of 87%. Again a higher sensitivity than parasitological techniques has been claimed, since some of the PCR positive persons were only confirmed parasitologically 4 to 6 months later.

Targeting the 18S rRNA gene, which is also a multi-copy gene and has sequences conserved within the *Trypanosomatidae* and within *Trypanozoon*, in the PCR-oligochromatography format allows detection of only 6 parasites/ml resulting in 100% sensitivity [33].

The above primers are all specific for the *Trypanozoon* subgenus, which excludes false positives due to other parasites such as *Plasmodium* or filaria, or transient infections with *T. congolense* or *T. vivax*, which are animal trypanosomes. Reported specificities are 98.6% with TBR and 97% with the ESAG primer sets, thus close to 100% [32, 34]. PCRs based on the above mentioned primers cannot discriminate between *T.b. gambiense* and *T.b. rhodesiense* infections, and might yield positive reactions with *T.b. brucei*, *T. evansi* and *T. equiperdum*. This led some authors to the hypothesis that the lack of specificity of the TBR based PCR in aparasitaemic CATT seropositives or in CATT negative subjects could be due to abortive *T.b. brucei* inoculations [35, 36]. Problems with TBR based PCR reproducibility and discordances with both sero-

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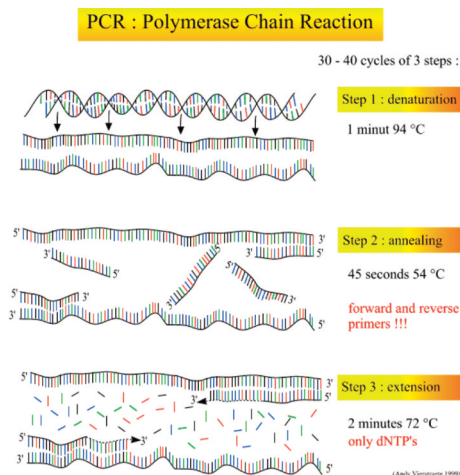
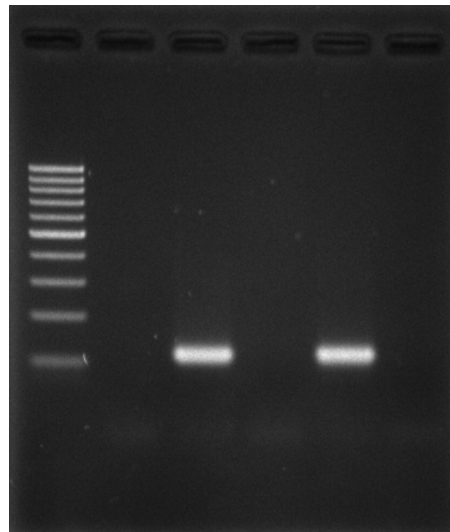
Fig 5a - principle of PCR reaction (picture from <http://users.ugent.be/~avierstr/principles/pcr.html>)

Fig 5b - Photo of stained agarose gel. The expected amplicon size is 110 bp.
Lane 1: DNA marker (100 bp ladder). Lane 2 & 4: negative sample. Lane 3: Positive sample.
Lane 5: positive control. Lane 6: negative control.

Figure 5 - The purpose of PCR is to make a large number of copies of a DNA fragment. This is done in 3 steps (figure 5a). 1° During the denaturation step at high temperature (94°C), the double DNA strand melts open to single stranded DNA. 2° During the annealing step (54°C) the single stranded primer binds to its corresponding single stranded DNA sequence. Primers that fit exactly bind more strongly on the DNA template. 3° During the extension step (72°C) the polymerase copies the DNA template. The polymerase starts copying at the site where primers annealed. Due to the higher temperature, primers that are on positions with no exact match, detach and don't give an extension. Step 1-3 are repeated 30-40 times. Because both strands are copied during PCR, there is an exponential increase of the number of copies (amplicons). Thereafter, the PCR product is visualized by agarose gel electrophoresis and ethidium bromide staining (figure 5b). The PCR reaction is positive when an amplicon has the expected size.

logical and parasitological results, especially in CATT seropositive subjects, were more recently attributed to the possibility of human reservoirs without detectable parasitaemia [29, 36]. In such subjects, the very low amount of circulating DNA would induce a random possibility of positive or negative PCR result when the test is repeated on the same sample [29].

Applications of PCR distinguishing *T.b. rhodesiense* or *T.b. gambiense* have been reported. For *T.b. rhodesiense*, these are based on the serum resistance-associated gene (SRA [37-39], for *T.b. gambiense* on the *T.b. gambiense* specific glycoprotein (TgsGP) gene [39, 40]. Both SRA and TgsGP are single copy genes thus limiting the sensitivity of derived PCR tests in case of low parasitaemia, which is commonly observed in *T.b. gambiense* HAT. Indeed, the detection limit of the TgsGP based PCR was only 1000 trypanosomes/ml after a single PCR run.

None of the published methods has been extensively validated for diagnosis, nor has any comparative study taken place. Therefore, much work has still to be done before molecular techniques can replace parasite detection in sleeping sickness, provided it ever will.

Nevertheless, some improvements in molecular diagnosis can be expected in the near future: i) simplification of sample preparation such as the use FTA paper for storage of blood [31, 39], ii) improved visualization and quantification of the PCR result [31, 33], iii) reduced need for expensive equipment, e.g. by using isothermal DNA amplification reactions such as in LAMP (Loop-mediated isothermal AMPLification test) or by using hybridization techniques [33, 41, 42].

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